



Sri Lanka Institute of Information Technology

IT3021 - Data warehousing and Business Intelligence

Assignment 02

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IT23302044

DS.WE.Y3.S1.01.02

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Step 1 — Data Source Description

1. Data Source for Assignment 2

The data source for Assignment 2 is the **Superstore Data Warehouse** (Superstore_DW) implemented in Assignment 1. This data warehouse was built on SQL Server 2025 and contains transformed, cleaned and aggregated data from the Superstore Sales dataset.

Data Warehouse Details:

Attribute	Detail
Database Name	Superstore_DW
Database Server	DESKTOP-0T1VSQE\MSSQLSERVER2
Schema Type	Snowflake Schema
Total Tables	8 (7 dimensions + 1 fact)
Total Records	9,986 fact records

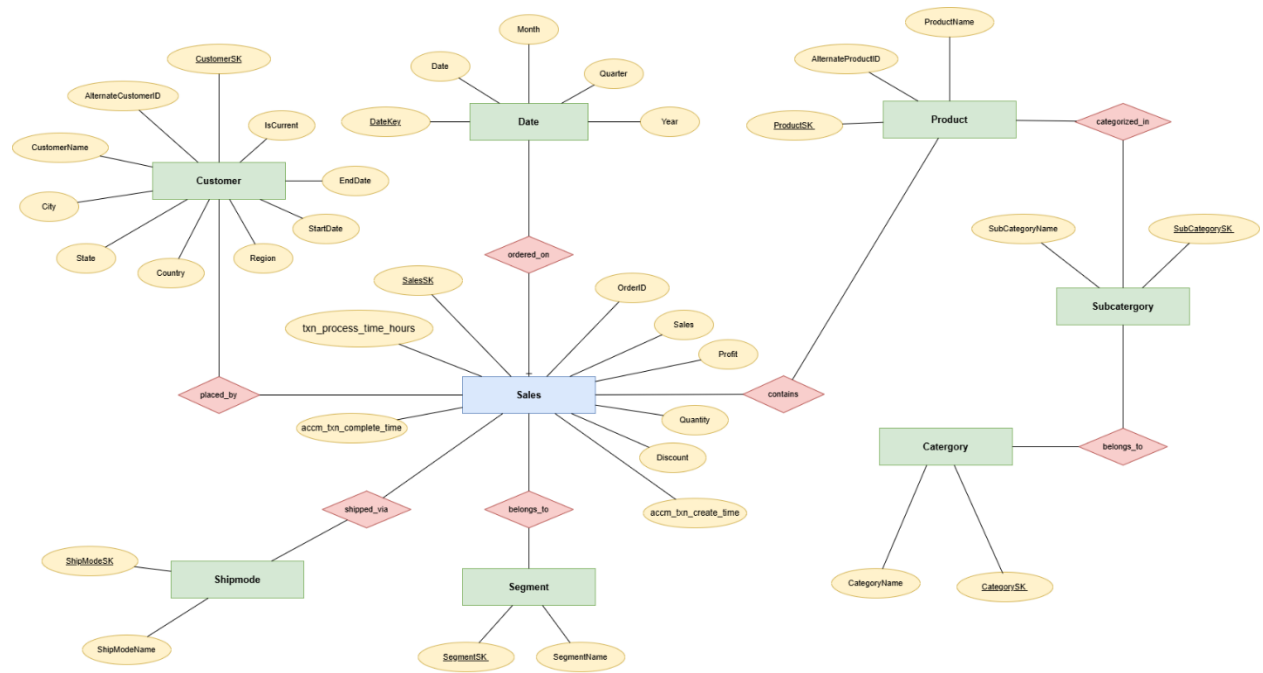
Dimension Tables:

Table	Rows	Description
DimCategory	3	Product categories
DimSubCategory	17	Product subcategories
DimProduct	1,862	Individual products
DimCustomer	793	Customer details (SCD Type 2)
DimShipMode	4	Shipping modes
DimSegment	3	Customer segments
DimDate	~39,000	Date dimension

Fact Table:

Table Rows Measures

Table	Rows	Measures
FactSales	9,986	Sales, Profit, Quantity, Discount, txn_process_time_hours



Step 2 — SSAS Cube Implementation

An SSAS (SQL Server Analysis Services) Multidimensional cube was created using SQL Server 2019 Analysis Services instance (DESKTOP-OT1VSQE\SSAS2019) with the Superstore Data Warehouse as the data source.

Step 1 — Created SSAS Project in Visual Studio

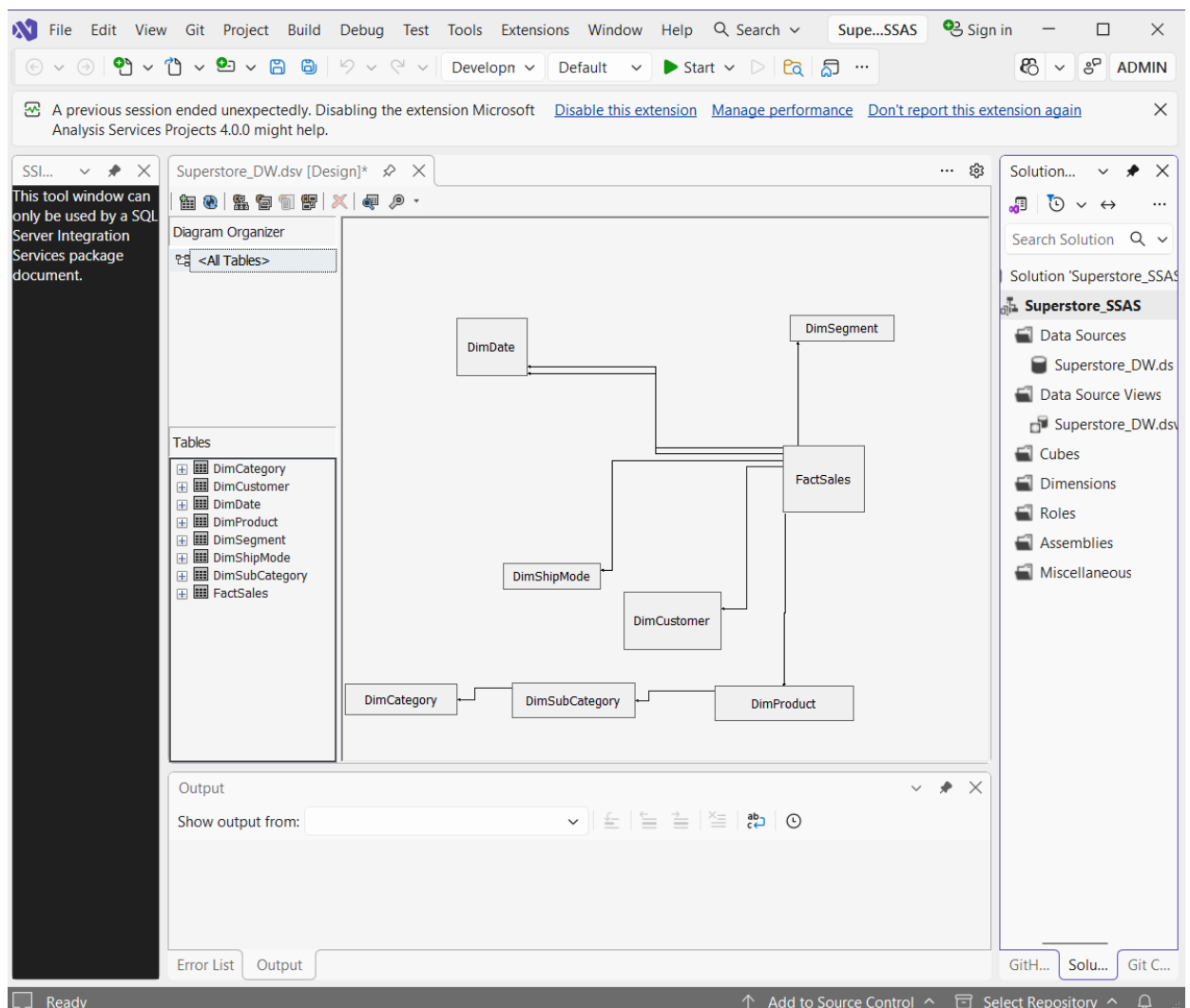
A new Analysis Services Multidimensional project named Superstore_SSAS was created in Visual Studio.

Step 2 — Created Data Source

A data source connection was created pointing to Superstore_DW database on DESKTOP-OT1VSQE\MSSQLSERVER2.

Step 3 — Created Data Source View

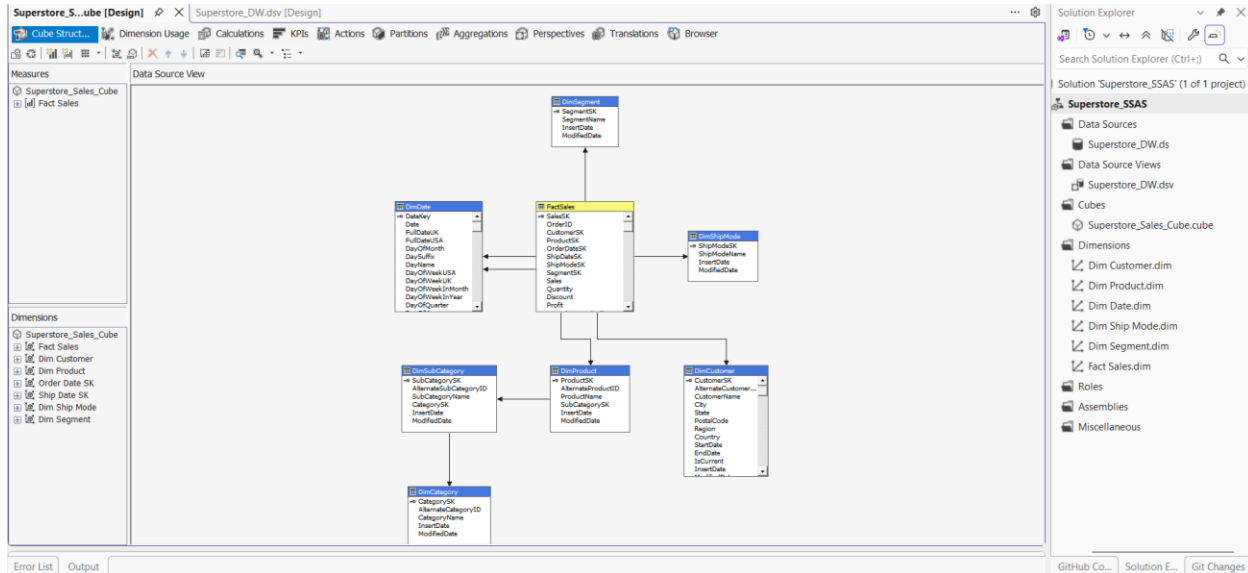
All 8 tables from the data warehouse were added to the Data Source View. Missing relationships were manually added to connect FactSales to all dimension tables.



Step 4 — Created the Cube

The Superstore_Sales_Cube was created using the Cube Wizard:

- **Measure Group:** FactSales
- **Measures included:** Sales, Profit, Quantity, Discount, Txn Process Time Hours
- **Dimensions included:** DimCustomer, DimProduct, DimSubCategory, DimCategory, DimDate, DimShipMode, DimSegment



Step 5 — Created Product Hierarchy

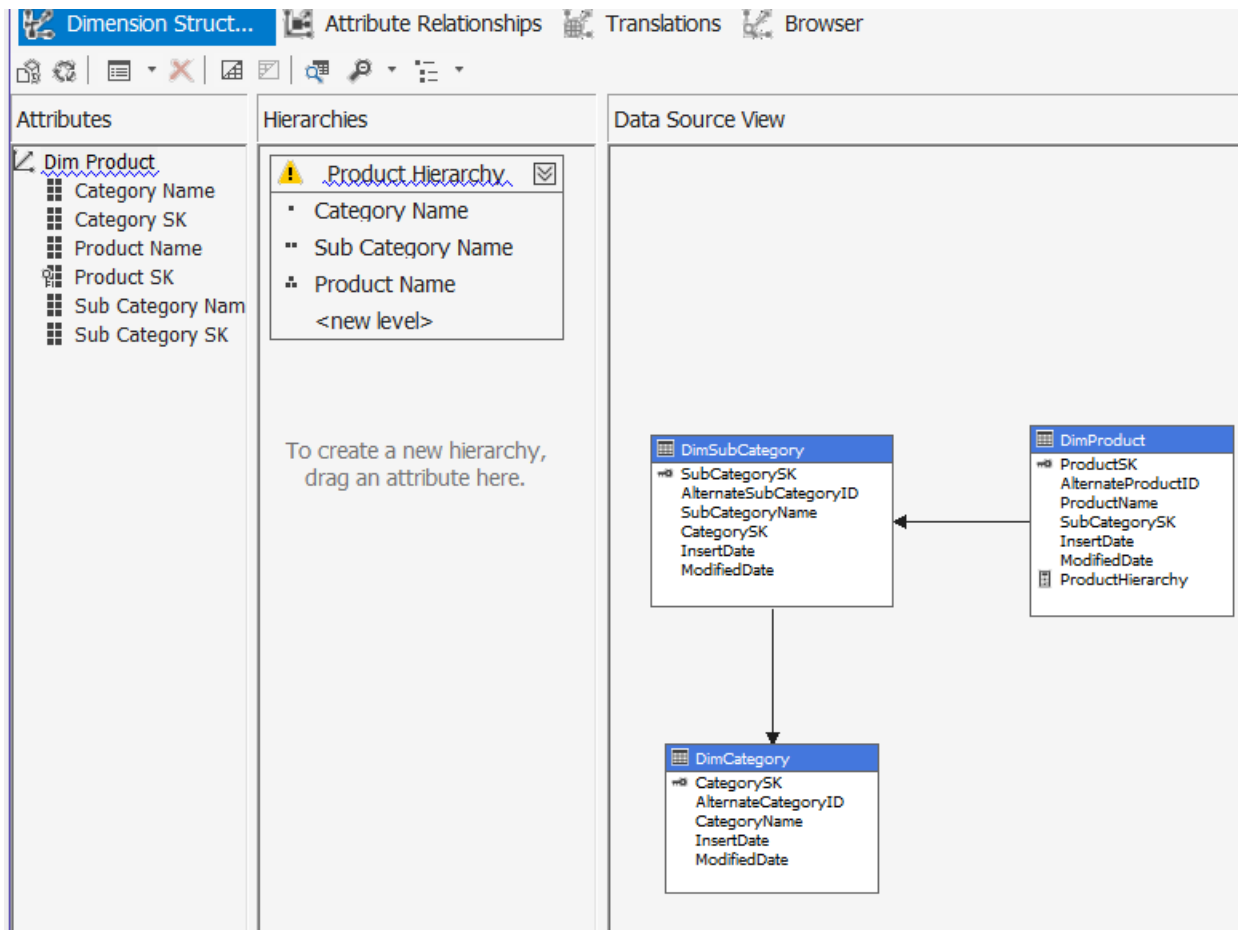
A three-level hierarchy was created in the DimProduct dimension:

Product Hierarchy

CategoryName - Level 1

SubCategoryName - Level 2

ProductName - Level 3



Step 6 — Created Date Hierarchy

A four-level date hierarchy was created in the DimDate dimension:

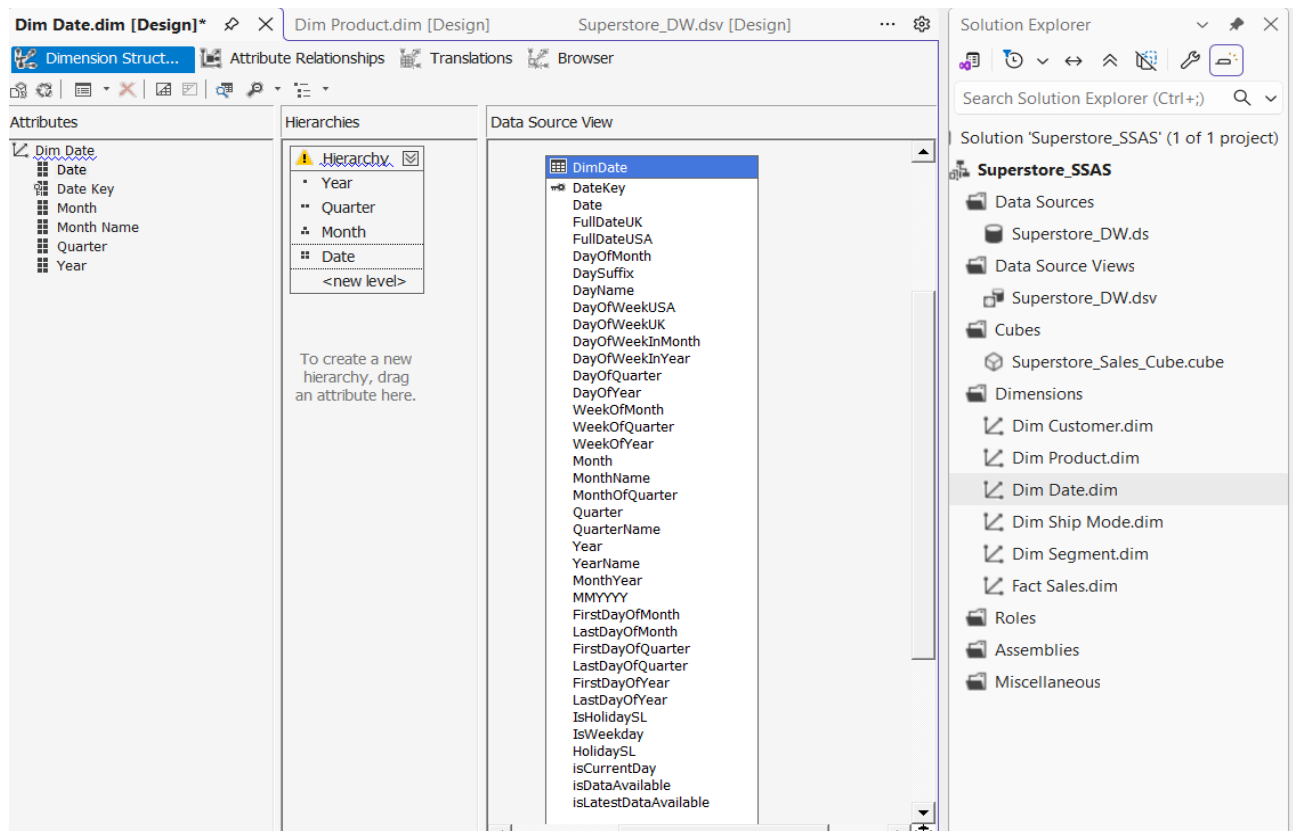
Date Hierarchy

Year - Level 1

Quarter - Level 2

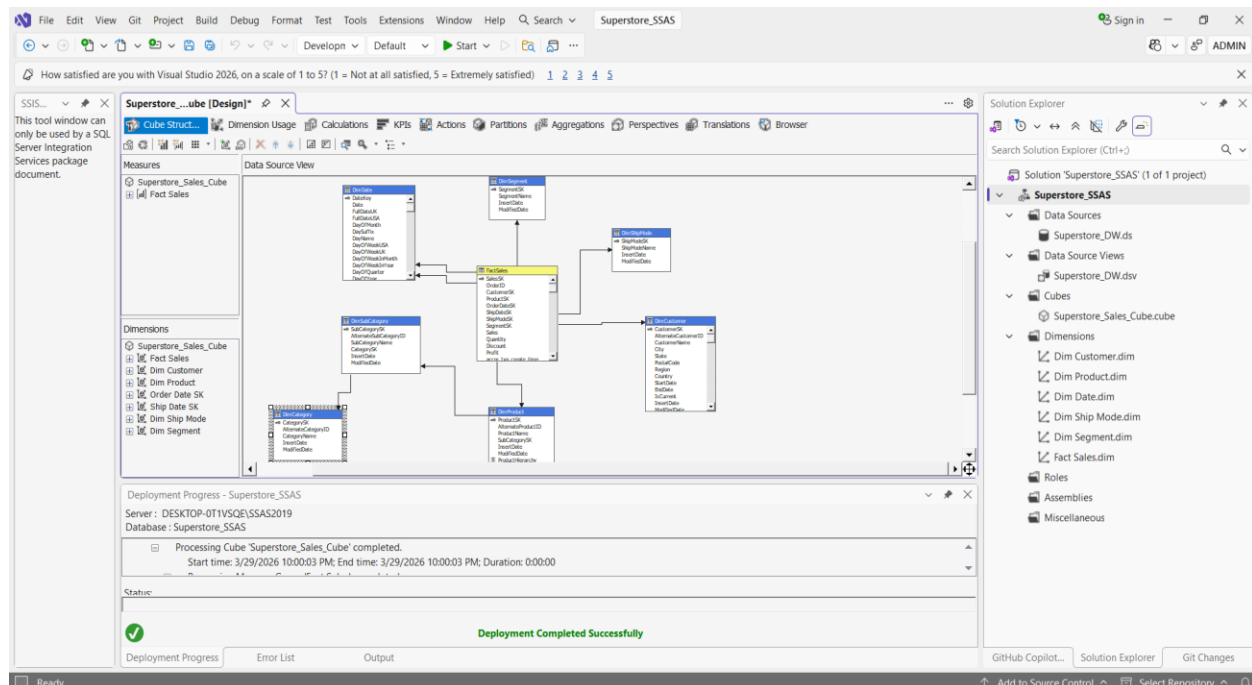
Month - Level 3

Date - Level 4

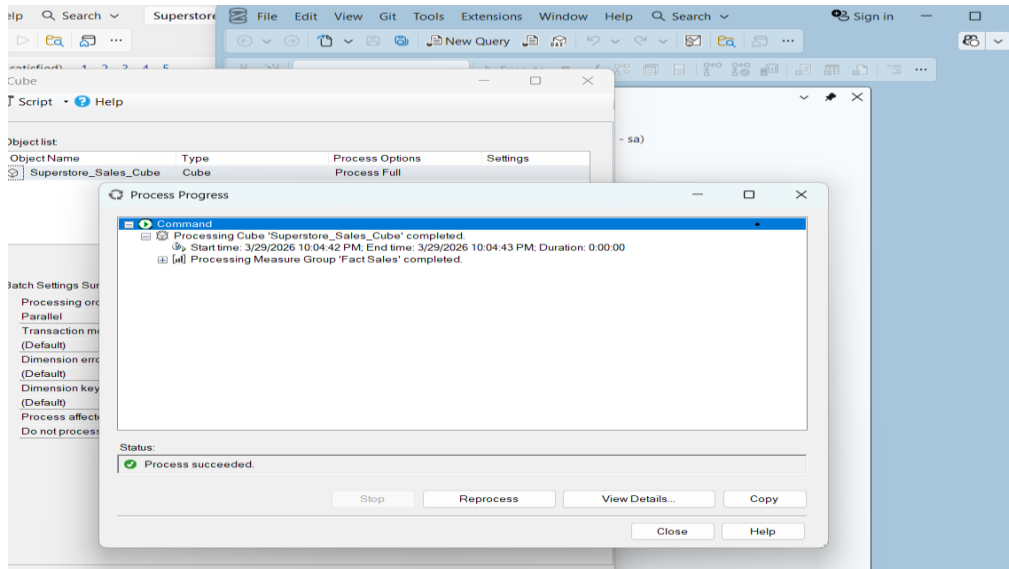


Step 7 — Deployed and Processed the Cube

The cube was deployed to DESKTOP-0T1VSQE\SSAS2019 and processed successfully. All data from the Superstore DW was loaded into the cube.



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2.3 Cube Structure Summary

Component	Details
Cube Name	Superstore_Sales_Cube
SSAS Instance	DESKTOP-0T1VSQE\SSAS2019
Measure Group	FactSales
Number of Measures	5
Number of Dimensions	7
Hierarchies	Product Hierarchy, Date Hierarchy

2.4 Measures in the Cube

Measure	Description	Aggregation
Sales	Total sales amount	Sum
Profit	Total profit amount	Sum
Quantity	Total quantity ordered	Sum
Discount	Total discount given	Sum
Txn Process Time Hours	Transaction processing time	Sum

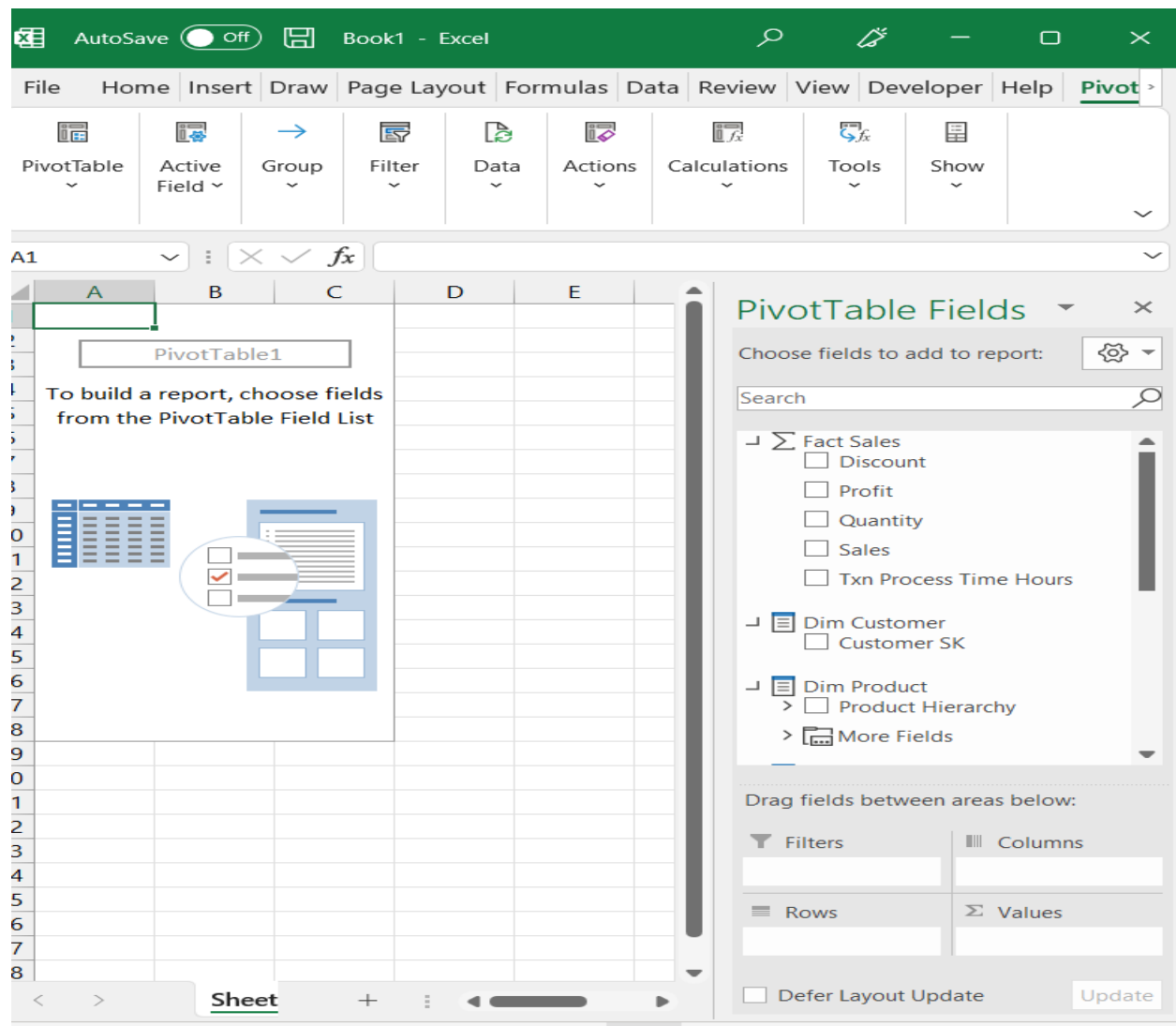
Step 3 — OLAP Operations in Excel

3. Demonstration of OLAP Operations

An Excel workbook was connected to the Superstore SSAS cube using the **Data → Get Data → From Analysis Services** feature. PivotTables were used to demonstrate all 5 OLAP operations.

Connection Details:

- **Server:** DESKTOP-OT1VSQE\SSAS2019
- **Database:** Superstore_SSAS
- **Cube:** Superstore_Sales_Cube
- **File:** Superstore_OLAP_Analysis.xlsx



3.1 Roll-Up

Roll-up aggregates data from a lower level to a higher level of a dimension hierarchy.

Steps followed:

- Dragged Sales to Values area
- Dragged CategoryName to Rows area
- Sales figures were aggregated at Category level (3 categories)

Result: Sales data was rolled up from individual product level to Category level showing total sales for Furniture, Office Supplies and Technology.

The screenshot shows the Microsoft Excel interface with the 'PivotTable Analyze' ribbon selected. The PivotTable is located in the range A1:B5. The PivotTable Fields task pane is open on the right side of the screen.

Row Labels	Sales
Furniture	741426.94
Office Supplies	718306.36
Technology	835756.22
Grand Total	2295489.5

The PivotTable Fields task pane shows the following configuration:

- Choose fields to add to report:** Search bar is present.
- Category Name** is selected in the list.
- More Fields** are listed: Category Name, Category SK, Product Name, Product SK.
- Drag fields between areas below:**
 - Filters:** (Empty)
 - Columns:** (Empty)
 - Rows:** Product Hier...
 - Values:** Sales
- Defer Layout Update:** (Unchecked)
- Update:** (Button)

3.2 Drill-Down

Drill-down navigates from a higher summary level to a lower detailed level.

Steps followed:

- Expanded CategoryName in Rows area
- Added SubCategoryName below CategoryName
- Each category was expanded to show its subcategories

Result: Furniture category drilled down to show Bookcases, Chairs, Furnishings and Tables with individual sales figures.

The screenshot displays the Microsoft Excel interface with a PivotTable and the PivotTable Fields task pane.

PivotTable Data:

Row Labels	Sales
Furniture	
Bookcases	114879.23
Chairs	328165.82
Furnishings	91417.29
Tables	206964.6
Office Supplies	
Appliances	107531.45
Art	27117.58
Binders	203318.02
Envelopes	16475.99
Fasteners	3023.94
Labels	12485.91
Paper	78382.18
Storage	223297.99
Supplies	46673.3
Technology	
Accessories	166985
Copiers	149527.85
Machines	189238.43
Phones	330004.94
Grand Total	2295489.52

PivotTable Fields Task Pane:

- Choose fields to add to report:** Search
- Dim Product**
 - ☒ **Product Hierarchy**
 - Category Name**
 - Sub Category Name**
 - Product Name
 - ☐ More Fields
 - ☐ Category Name
 - ☐ Category SK

Drag fields between areas below:

Filters	Columns
Rows	Values
Product Hier...	Sales

☐ Defer Layout Update Update

3.3 Slice

Slice filters the data by selecting a single value from one dimension.

Steps followed:

- Dragged Region from DimCustomer to Filters area
- Selected **West** as the filter value
- PivotTable updated to show only West region data

Result: Data was sliced to show only West region sales across all categories and subcategories.

The screenshot displays the Microsoft Excel interface with a PivotTable and the PivotTable Fields task pane.

Excel Ribbon: File, Home, Insert, Draw, Page Layout, Formulas, Data, Review, View, Developer, Help, PivotTable Analyze, Design.

Formulas Ribbon: Undo, Clipboard, Font, Alignment, Number, Conditional Formatting, Format as Table, Cell Styles, Styles, Cells, Editing, Analyze Data, Sensitivity.

Formula Bar: B4, X, Y, fx

PivotTable Data:

Region	West
Row Labels	Sales
Furniture	
Bookcases	37023.61
Chairs	122640.64
Furnishings	32431.71
Tables	87177.23
Office Supplies	
Appliances	37651.3
Art	8720.62
Binders	80526.17
Envelopes	4326.11
Fasteners	980.64
Labels	5338.1
Paper	23991.34
Storage	64927.65
Supplies	13609.76
Technology	
Accessories	57009.74
Copiers	41359.42
Machines	38524.28
Phones	108287.38
Grand Total	764525.7

PivotTable Fields Task Pane:

- Choose fields to add to report: [Settings]
- Search: [Search]
- Available Fields: City, Country, Customer Name, Customer SK, Postal Code, **Region** (checked), State.
- Drag fields between areas below:
 - Filters:** Region
 - Columns:**
 - Rows:** Product Hier...
 - Values:** Sales
- Defer Layout Update: [] Update

3.4 Dice

Dice filters the data by selecting values from multiple dimensions simultaneously.

Steps followed:

- Kept Region = West filter
- Added SegmentName from DimSegment to Filters area
- Selected **Consumer** as the segment filter

Result: Data was diced to show only West region AND Consumer segment sales — two dimensions filtered simultaneously.

The screenshot shows an Excel spreadsheet with a PivotTable and the PivotTable Fields task pane. The PivotTable is located in the range B5:F25 and displays sales data for the West region, filtered by Consumer segment. The PivotTable Fields task pane is on the right, showing the following configuration:

- Filters:** Region, Segment Name
- Columns:** (Empty)
- Rows:** Product Hier...
- Values:** Sales

The PivotTable data is as follows:

Row Labels	Sales
Furniture	
Bookcases	23142.72
Chairs	75018.62
Furnishings	20992.22
Tables	47295.11
Office Supplies	
Appliances	23311.13
Art	5139.94
Binders	52851.41
Envelopes	1953.53
Fasteners	579.76
Labels	2969.53
Paper	12177.77
Storage	33200.56
Supplies	6601.19
Technology	
Accessories	33340.34
Copiers	15679.68
Machines	28196.99
Phones	62178.97
Grand Total	444629.47

3.5 Pivot

Pivot rotates the data axes to view data from a different perspective.

Steps followed:

- Moved CategoryName from Rows area to Columns area
- Dragged ShipModeName from DimShipMode to Rows area

Result: The PivotTable was rotated showing Ship Modes in rows and Categories in columns — a completely different perspective of the same data.

The screenshot displays the Microsoft Excel interface with the PivotTable Fields task pane open on the right. The PivotTable is located in the worksheet 'Sales' and shows data categorized by Ship Mode (rows) and Product Hierarchy (columns). The data is summarized by Sales.

Ship Mode	Furniture	Office Supplies	Technology	Grand Total
First Class	110729.7	101281.44	139365.03	351376.17
Same Day	39148.44	29089.87	60123.68	128361.99
Second Class	156001.67	160136.74	142118.54	458256.95
Standard Class	435547.13	427798.31	494148.97	1357494.4
Grand Total	741426.94	718306.36	835756.22	2295489.5

The PivotTable Fields task pane on the right shows the following configuration:

- Product Hierarchy** (checked): Category Name, Sub Category Name, Product Name, More Fields
- Dim Segment** (checked): Segment Name, Segment SK
- Filters**: (Empty)
- Columns**: Product Hiera...
- Rows**: Ship Mode ...
- Values**: Sales
- Defer Layout Update**: (unchecked)
- Update**: (button)

Summary of OLAP Operations:

Operation	Description	Dimension Used
Roll-Up	Aggregated sales by Category	DimProduct (Category level)
Drill-Down	Expanded Category to SubCategory	DimProduct Hierarchy
Slice	Filtered by Region = West	DimCustomer (Region)
Dice	Filtered by Region = West AND Segment = Consumer	DimCustomer + DimSegment
Pivot	Rotated Categories to Columns, Ship Mode to Rows	DimProduct + DimShipMode

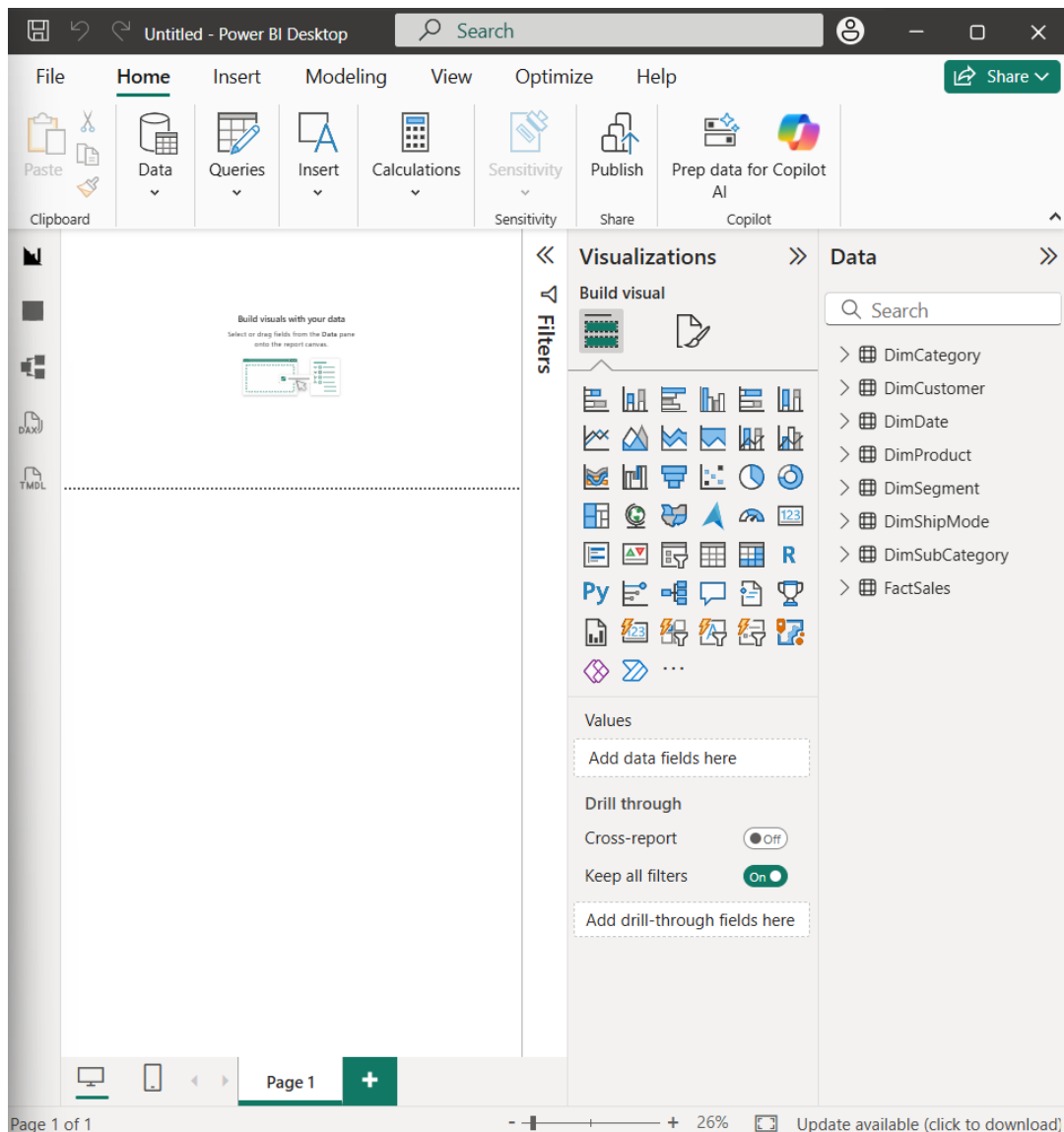
Step 4 — Power BI Reports

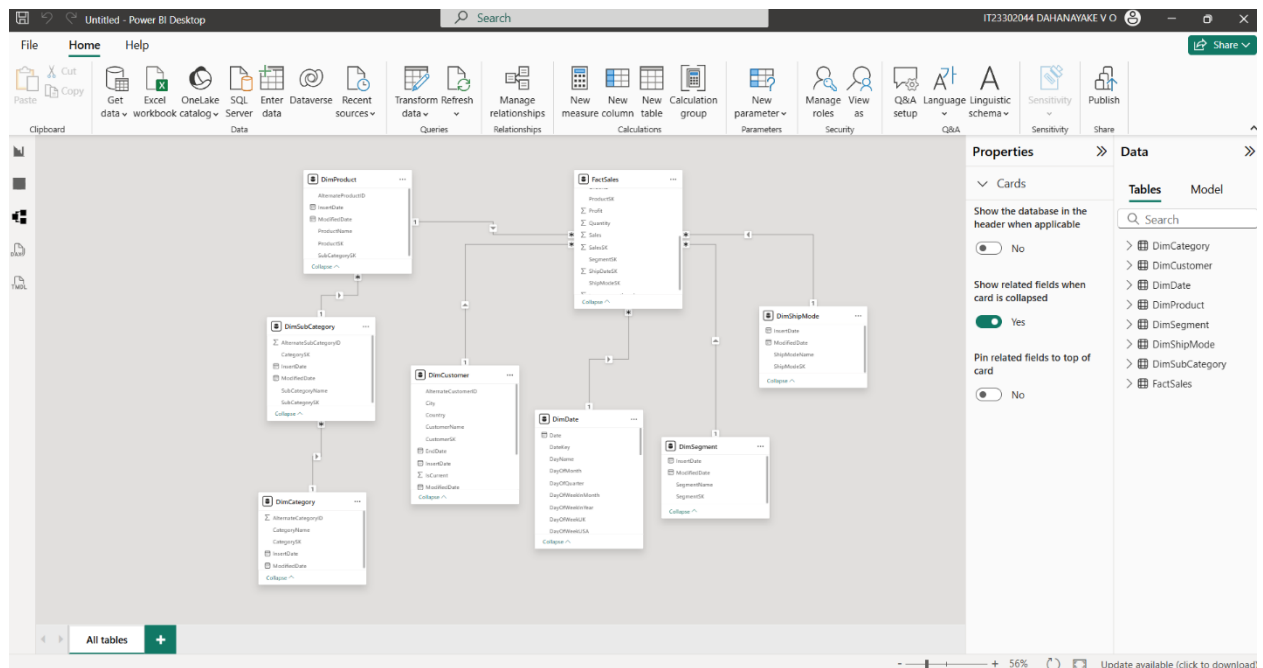
4. Power BI Reports

Four Power BI reports were developed and published to Power BI Service using **Superstore_DW** as the data source. All reports were created in Power BI Desktop and published to **My Workspace** in Power BI Service.

Connection Details:

- **Server:** DESKTOP-OT1VSQE\MSSQLSERVER2
- **Database:** Superstore_DW
- **Connectivity Mode:** Import
- **Published to:** Power BI Service — My Workspace





4.1 Report 1 — Matrix Visual

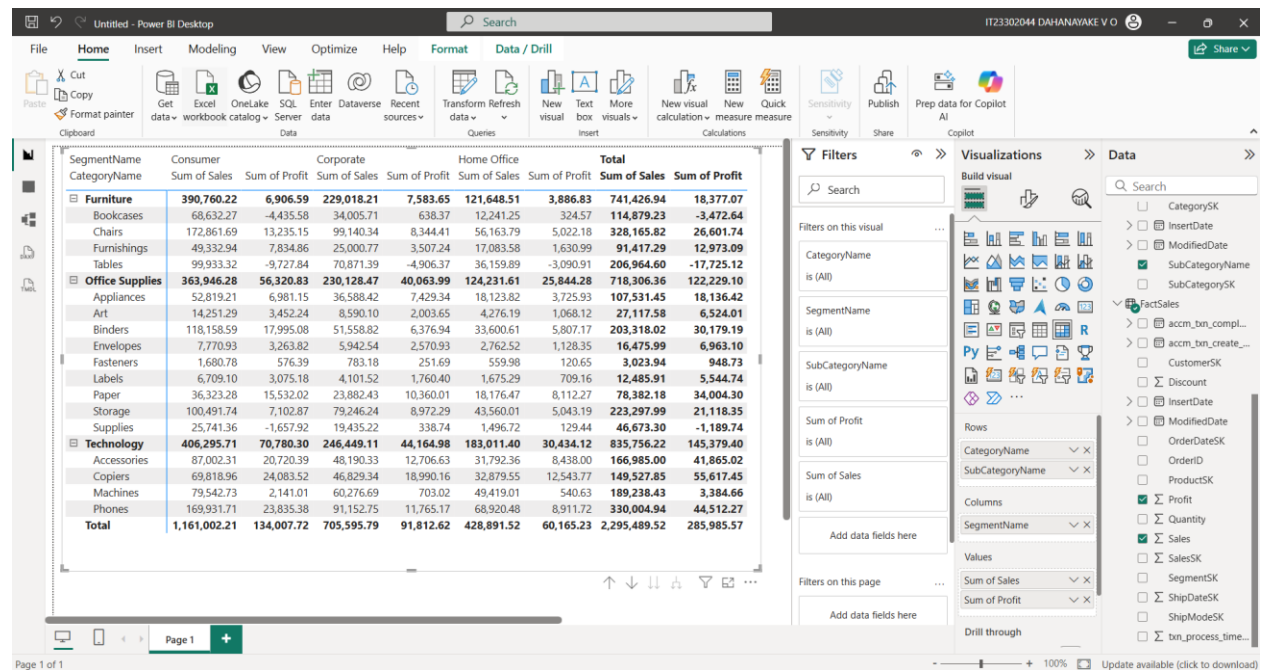
Objective:

Display detailed tabular data with row and column groupings.

Steps followed:

1. Added a Matrix visual to the report canvas
2. Configured the Matrix:
 - **Rows:** CategoryName → SubCategoryName (hierarchy)
 - **Columns:** SegmentName
 - **Values:** Sales, Profit

Result: The Matrix visual displays Sales and Profit broken down by Category and SubCategory in rows, and Customer Segments (Consumer, Corporate, Home Office) in columns — allowing easy comparison across segments and product hierarchy.



4.2 Report 2 — Cascading Slicers

Objective:

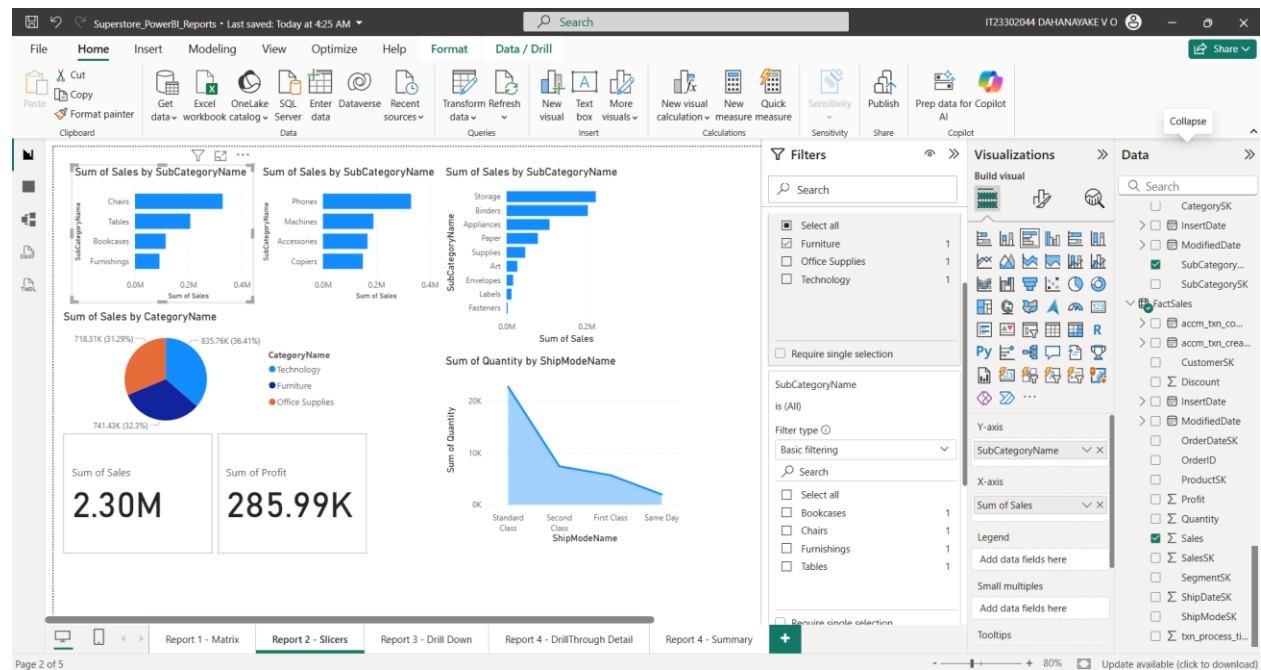
Create multiple slicers where selecting a value in the first slicer dynamically filters the options in the second slicer.

Steps followed:

1. Added a **List Slicer** for CategoryName
2. Added a **List Slicer** for SubCategoryName
3. Added a **Clustered Bar Chart** showing Sales by SubCategory
4. Configured slicer interactions so Category slicer filters SubCategory slicer

Result:

When **Furniture** is selected in the Category slicer, the SubCategory slicer automatically shows only Furniture subcategories (Bookcases, Chairs, Furnishings, Tables) and the bar chart updates accordingly — demonstrating cascading filters.



4.3 Report 3 — Drill-Down Report

Objective:

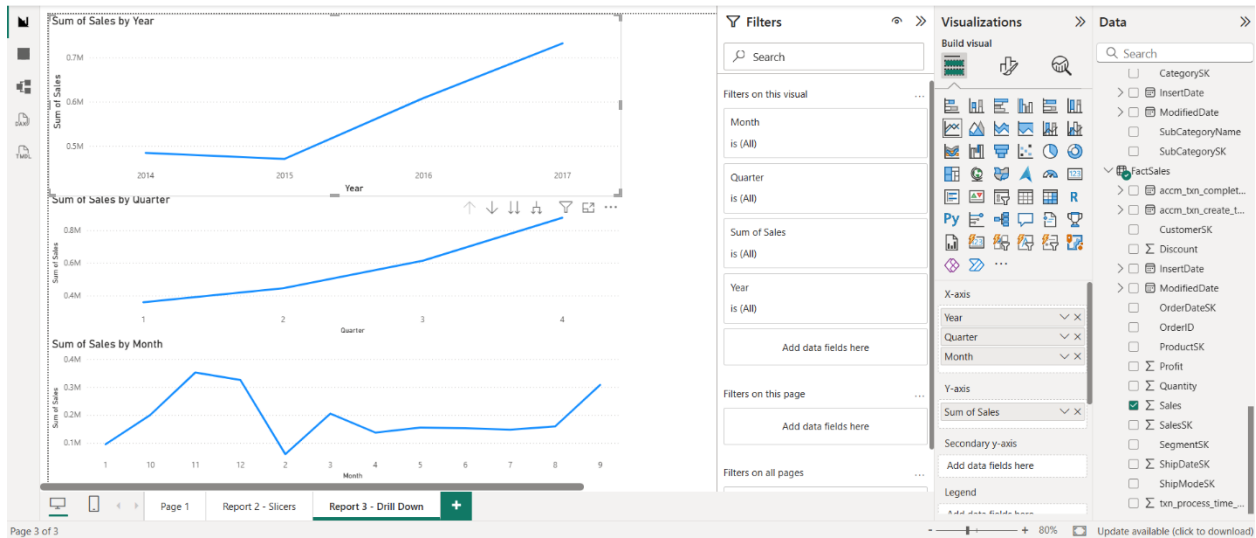
Allow users to explore data hierarchically from Year to Quarter to Month.

Steps followed:

1. Added a **Clustered Bar Chart**
2. Configured:
 - **X-axis:** Year → Quarter → Month (from DimDate)
 - **Y-axis:** Sales
3. Enabled drill-down using the drill-down arrows on the chart

Result:

Users can click the drill-down arrow to navigate from Year level → Quarter level → Month level. Clicking the drill-up arrow returns to the previous level — demonstrating hierarchical data exploration.



4.4 Report 4 — Drill-Through Report

Objective:

Allow users to right-click on a visual and navigate to a detailed page showing related information.

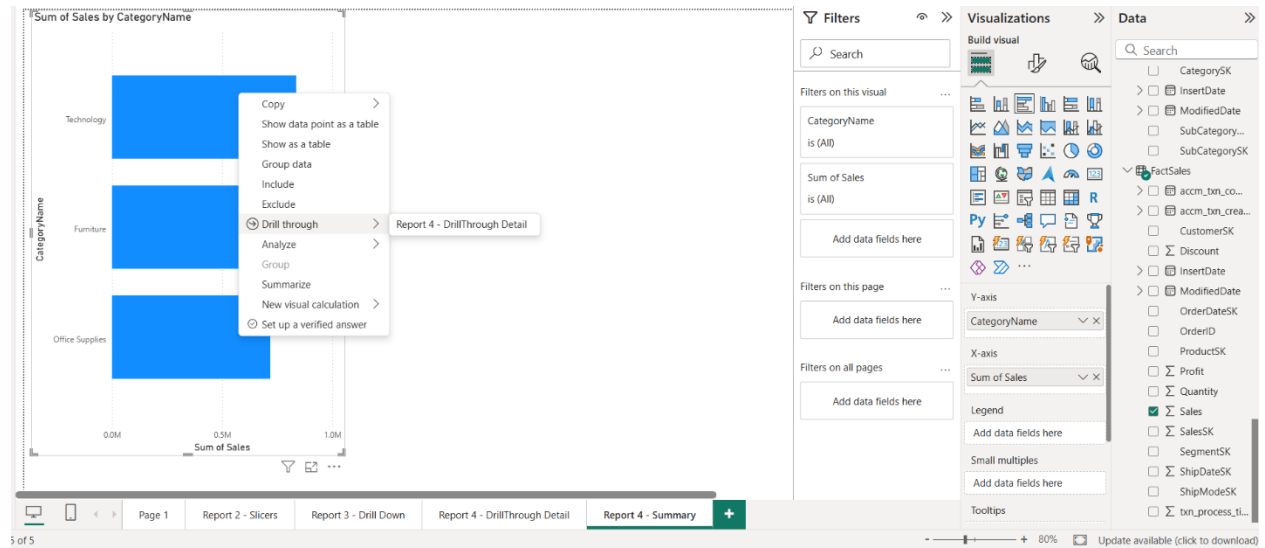
Steps followed:

1. Created a **Summary page** with a Bar Chart showing Sales by Category
2. Created a **Detail page** with a Table showing:
 - OrderID, CustomerName, ProductName, Sales, Profit, Quantity
3. Added CategoryName as the **Drill-through field** on the detail page

Result:

Users can right-click on any category bar in the Summary page → select **Drill through** → navigate to the Detail page filtered by that specific category — showing all individual orders for that category.

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OrderID	CustomerName	ProductName	Sum of Sales	Sum of Profit	Sum of Quantity
CA-2014-100006	Dennis Kane	AT&T EL31110 DECT	377.97	109.61	3
CA-2014-100076	Kunsti Miller	Remington Expert Mouse Optical USB Trackball for PC or Mac	227.97	28.49	3
CA-2014-100706	Laurel Elliott	Case Logic 2.4GHz Wireless Keyboard	99.98	7.99	2
CA-2014-100867	Eugene Hildebrand	RCA Viqix Integrated PBX 8-Line Router	321.55	20.09	6
CA-2014-100881	Daniel Raglin	AT&T TR1909W	302.37	22.67	3
CA-2014-100895	Stewart Visinsky	Mophie Juice Pack Helium for iPhone	239.97	67.19	3
CA-2014-101476	Shirley Daniels	"Epson WorkForce WF-2530 All-in-One Printer, Copier Scanner"	69.99	30.09	1
CA-2014-101602	Mick Crebaggia	ARKON Windshield Dashboard Air Vent Car Mount Holder	40.68	-9.15	3
CA-2014-102274	Dave Hallsten	"Logitech B530 USB Headset - headset - Full size, Binaural"	36.99	9.98	1
CA-2014-102330	Arianne Irving	HTC One	319.96	35.99	4
CA-2014-102673	Ken Heidel	Cisco IP Phone 7961G VoIP phone - Dark gray	978.84	110.11	9
CA-2014-102869	Logan Currie	Polycom SoundPoint IP 450 VoIP phone	135.51	-31.62	1
CA-2014-102869	Logan Currie	Rosewill 107 Normal Keys USB Wired Standard Keyboard	64.70	-4.85	6
CA-2014-102988	Greg Maxwell	AT&T CL82213	28.99	8.40	1
CA-2014-103058	Alejandro Grove	Case Logic 2.4GHz Wireless Keyboard	99.98	7.99	2
CA-2014-103310	Greg Matthias	Wilson Electronics DB Pro Signal Booster	1,432.00	125.30	5
CA-2014-103366	Erica Hackney	Logitech 910-002974 M325 Wireless Mouse for Web Scrolling	149.95	65.97	5
CA-2014-103373	Bruce Stewart	Apple iPhone 5	779.79	-168.95	2
CA-2014-103429	Laurel Workman	Geemarc AmpliPOWER60	464.00	134.56	5
CA-2014-103492	Craig Molinari	Cisco 888 Inc. 6753i IP Business Phone System	755.94	66.14	7
CA-2014-103492	Craig Molinari	Motorola Droid Maxx	719.95	71.99	6
CA-2014-103660	Mich Webber	Polycom VVX 310 VoIP phone	1,007.94	75.59	7
CA-2014-103807	Elizabeth Moffitt	Kingston Digital DataTraveler 8GB USB 2.0	5.95	0.83	1
CA-2014-103849	Paul Gonzalez	SanDisk Cruzer 64 GB USB Flash Drive	58.11	7.26	2
CA-2014-103849	Paul Gonzalez	XBlue XB-1670-88 X16 Small Office Telephone - Titanium	100.79	6.29	1
CA-2014-103989	Matt Connell	"Logitech Wireless Boombox Speaker - portable - wireless, wired"	170.24	53.20	2
CA-2014-103989	Matt Connell	Cisco 888 Inc. 6753i IP Business Phone System	323.97	28.34	3
CA-2014-104178	Janet Molinari	SanDisk Cruzer 32 GB USB Flash Drive	95.10	30.43	5
CA-2014-104283	Liz Mackendrick	Sony Micro Vault Click 16 GB USB 2.0 Flash Drive	223.96	53.75	4
CA-2014-104402	Duane Benoit	Texas Instruments TI-34 Scientific Calculator	65.97	31.00	3
CA-2014-104738	Stefania Perino	"Plantronics CS 50-USB - headset - Convertible, Monoaural"	217.58	19.03	2
Total			835,756.22	145,379.40	6926

Summary of Power BI Reports:

Report	Visual Type	Key Feature
Report 1 — Matrix	Matrix	Row and column groupings with hierarchy
Report 2 — Slicers	Slicers + Bar Chart	Cascading filters between Category and SubCategory
Report 3 — Drill Down	Bar Chart	Year → Quarter → Month hierarchy navigation
Report 4 — Drill Through	Bar Chart + Table	Right-click navigation to detailed order data

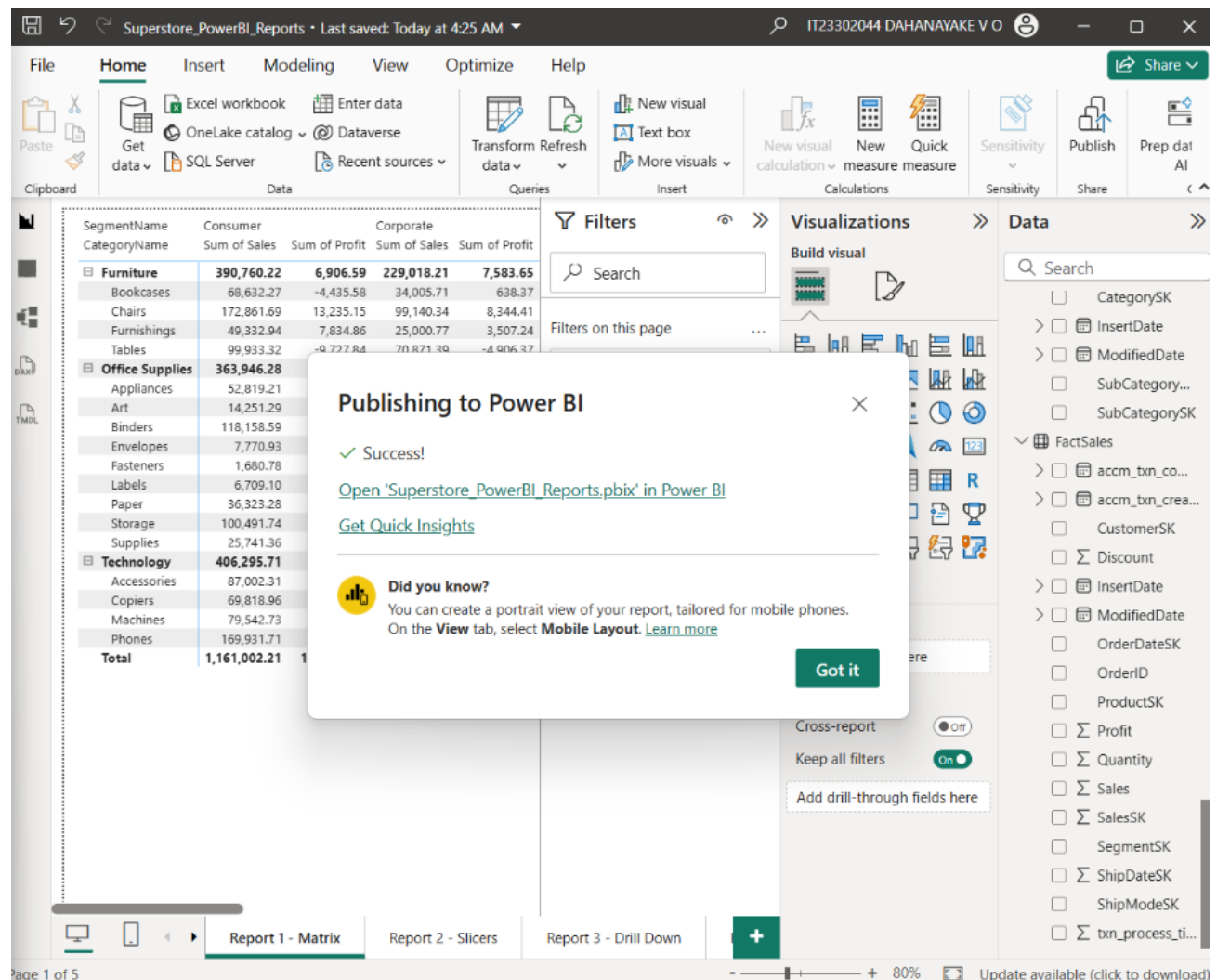
4.5 Publishing to Power BI Service

All four reports were published to **Power BI Service** using the **Publish** feature in Power BI Desktop. The reports were published to **My Workspace**.

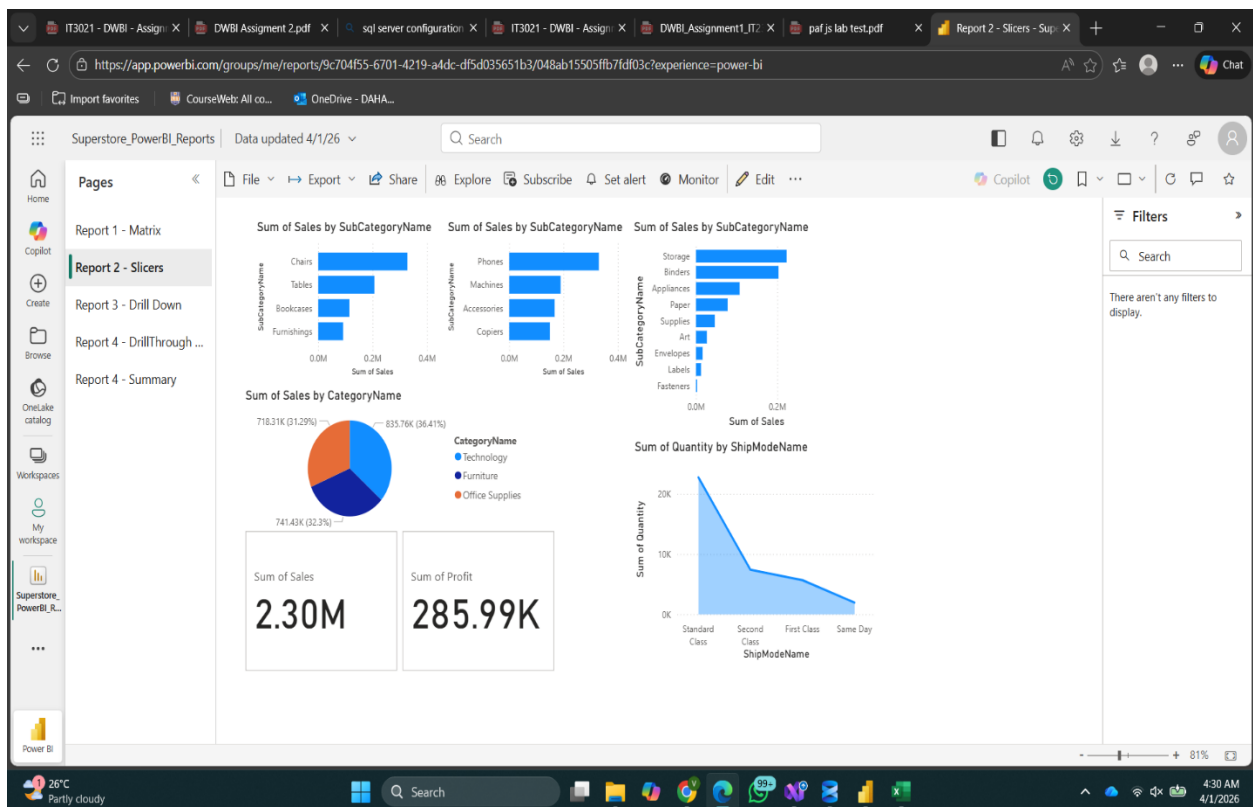
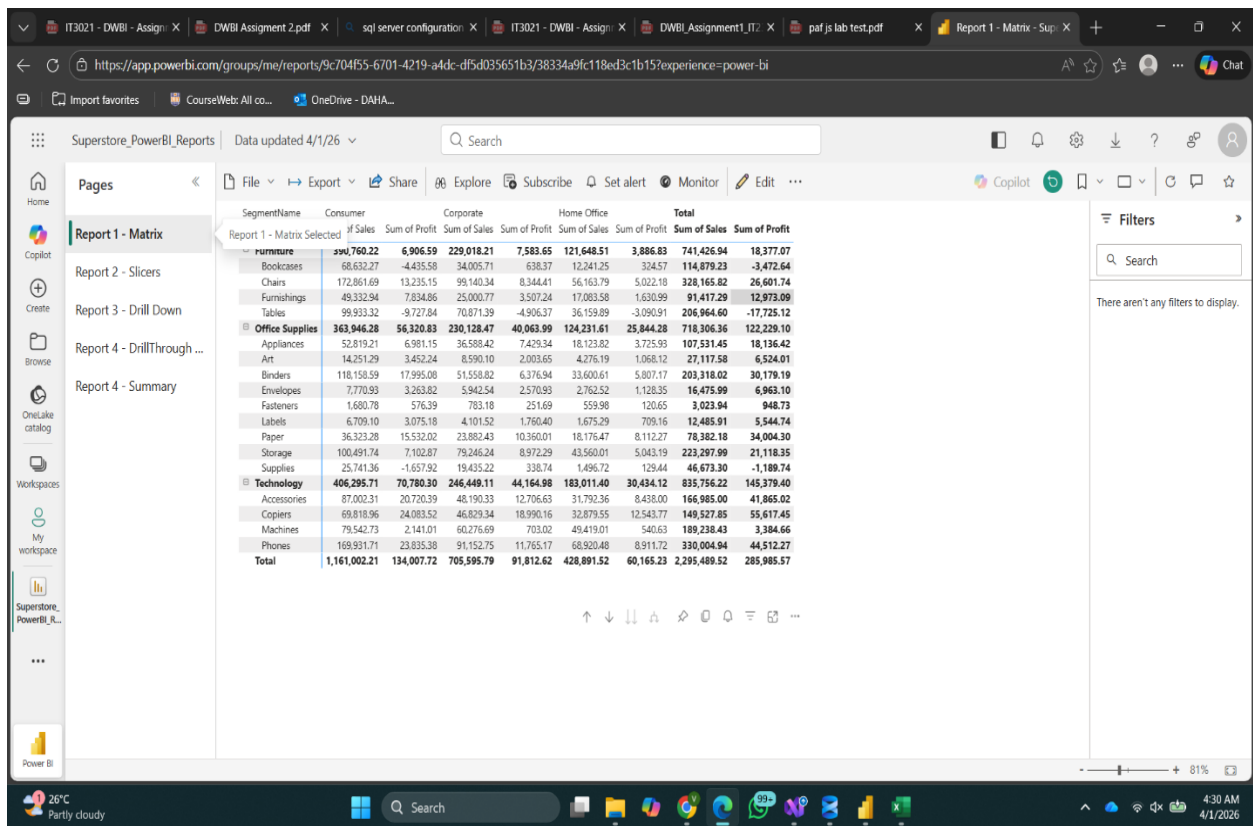
Publishing Steps followed:

1. Clicked **Home** → **Publish** in Power BI Desktop
2. Selected **My Workspace** as the destination
3. Waited for publishing to complete successfully
4. Opened each report in Power BI Service to verify they display correctly online

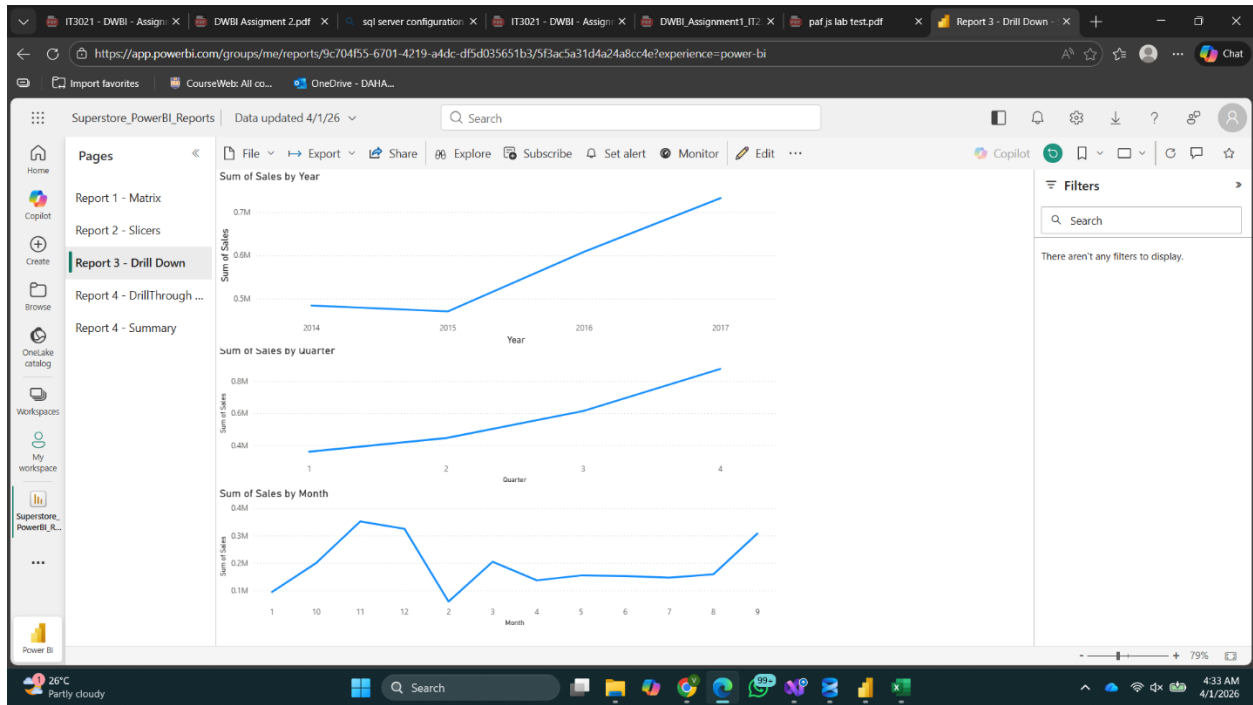
Result: All 4 reports are now accessible in Power BI Service and can be viewed, shared and interacted with online — demonstrating full end-to-end BI solution from data warehouse to online reporting.



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The screenshot shows the Microsoft Power BI Desktop interface. The main view displays a table of sales data with columns: OrderID, CustomerName, ProductName, Sum of Sales, Sum of Profit, and Sum of Quantity. The left sidebar shows the 'Pages' pane with 'Report 4 - DrillThrough...' selected. The top navigation bar includes 'File', 'Export', 'Share', 'Explore', 'Subscribe', 'Set alert', 'Monitor', and 'Edit'. The bottom status bar shows the temperature as 26°C and the time as 4:31 AM on 4/1/2026.

OrderID	CustomerName	ProductName	Sum of Sales	Sum of Profit	Sum of Quantity
CA-2014-100090	Ed Braton	Hon 2111 Invitation Series Corner Table	502.48	-87.93	3
CA-2014-100078	Kunst Miller	DWI Arturo Collection Mission-style Design Wood Chair	317.05	-18.11	3
CA-2014-100706	Laurel Elliston	Ultra Door Push Plate	29.46	9.72	6
CA-2014-100916	Frank Hawley	Hon 2111 Invitation Series Straight Table	591.32	112.35	4
CA-2014-101462	Benjamin Patterson	"GE 4 Foot Fluorescent Tube, 40 Watt"	59.92	27.56	4
CA-2014-101560	Chris Selesnick	Eidon Cleatmat Plus Chair Mats for High Pile Carpets	397.60	43.73	5
CA-2014-101560	Chris Selesnick	"Lifetime Advantage Folding Chairs, 4/Carton"	763.28	-21.80	5
CA-2014-102009	O'Sullivan Living	Dimensions 2 Shelf Bookcase	816.99	-35.29	6
CA-2014-102009	O'Sullivan Living	"Global Leather & Oak Executive Chair, Burgundy"	120.71	-18.10	1
CA-2014-102652	Andy Yotov	C-Line Cubicle Keepers Polypropylene Holder With Velcro Backings	33.11	12.91	7
CA-2014-102652	Andy Yotov	Tenex B1-RE Series Chair Mats for Low Pile Carpets	91.96	15.63	2
CA-2014-102869	Logan Currie	"Master Caster Door Stop, Large Neon Orange"	17.47	5.02	3
CA-2014-103086	Evan Baillet	G.E. Longer-Life Indoor Recessed Floodlight Bulbs	5.31	-1.59	2
CA-2014-103219	Benjamin Farhat	"Howard Miller 13" Diameter Goldtone Round Wall Clock"	56.52	21.47	9
CA-2014-103310	Greg Matthias	Hon Pagoda Stacking Chairs	256.78	32.09	1
CA-2014-103317	Don Miller	Advantus Panel Wall Certificate Holder - 8.5x11	19.52	5.36	2
CA-2014-103317	Don Miller	Luxo Adjustable Task Clamp Lamp	213.21	15.99	3
CA-2014-103331	Karl Braun	Acrylic Self-Standing Desk Frames	10.68	4.05	4
CA-2014-103849	Paul Gonzalez	"Deflect-o-EconoMat Studded, No Bevel Mat for Low Pile Carpeting"	66.11	-84.29	4
CA-2014-103940	Bradley Nguyen	"Executive Impressions 12" Wall Clock"	35.34	13.42	2
CA-2014-103989	Matt Connell	Document Clip Frames	20.01	5.50	3
CA-2014-103989	Matt Connell	Eidon Wave Desk Accessories	4.99	1.37	3
CA-2014-104178	Janet Molinari	"Eidon 200 Class Desk Accessories, Smoke"	6.28	2.63	1
CA-2014-104269	Dave Brooks	Global Deluxe High-Back Manager's Chair	457.56	51.47	2
CA-2014-104283	Liz MacKendrick	KI Adjustable-Height Table	85.98	22.35	1
CA-2014-104472	Christine Kargatis	Aluminum Document Frame	73.32	21.99	6
CA-2014-104563	Craig Molinari	"Global Leather and Oak Executive Chair, Black"	481.56	54.17	2
CA-2014-104563	Craig Molinari	Office Star - Task Chair with Contemporary Loop Arms	436.70	21.83	6
CA-2014-105165	Sam Zeldin	"Revere Furniture Oval Coffee Table, Oval End Table, End Table with Drawer"	200.79	-22.94	1
CA-2014-105249	Duane Huffman	"Sauder Camden County Barrister Bookcase, Planked Cherry Finish"	411.33	-4.83	4
Total			741,426.94	18,377.07	8023

DWBI ASSIGNMENT 2 – IT23302044

